# TerraBit: 1-bit Is All You Need for Earth Embedding Retrieval

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## Abstract

Geospatial foundation models now produce billion-scale embedding corpora whose storage and retrieval costs dominate deployment. TerraBit asks a practical question: how far can we quantize 1024D Earth embeddings while preserving nearest-neighbor structure needed for retrieval? We present a reproducible pipeline that combines intrinsic-dimension diagnostics, label-free fidelity metrics, and a parallel Parquet transformation system that preserves dataset schema and metadata while replacing only the embedding column. On $49\,785\,116$ real embeddings across $1021$ files, we show a smooth compression-fidelity frontier from $2\times$ (float16) to $16.53\times$ (1-bit binary) and complete full-corpus multi-method quantization in $45.2\,\mathrm{min}$ at $18\,358.77$ rows/s.

## 1 Introduction

High-dimensional geospatial embeddings are useful for retrieval, deduplication, change analysis, and indexing, but they are expensive to move, store, and search at scale. In this setting, every byte per vector matters.

TerraBit focuses on the deployment regime where embeddings are already produced, and the bottleneck is representation cost rather than model training. Our central claim is operational: *1-bit embeddings can be useful for high-recall shortlist retrieval, while higher-bit variants (int4/int8) provide strong quality at substantial compression.*

We make three contributions.

- A practical evaluation framework that combines intrinsic-dimension (ID) estimates with label-free retrieval and geometry metrics.
- A quantization suite spanning float16, fp8, int8, int4, int3, int2, and binary (1-bit sign) under a common interface.
- A production-oriented Parquet transformer that preserves non-embedding columns and schema metadata, writes quantization metadata into Parquet footers, and runs in parallel over large partitioned datasets.

This paper is intentionally short and systems-oriented: we report concrete trade-offs, reproducible scripts, and limitations.

## 2 Related work

TerraBit sits at the intersection of three threads. First, dimensionality reduction and intrinsic-dimension analysis motivate that many high-dimensional representations have lower effective structure than their ambient dimension. Second, vector quantization and low-bit compression are common in large-scale retrieval systems because memory bandwidth and cache behavior often bound latency. Third, geospatial retrieval pipelines typically require preserving spatial metadata and storage layouts, not only vector fidelity.

Our emphasis differs from model-centric compression work: we study post-hoc quantization of an existing, real embedding lake, and evaluate quality-cost trade-offs with label-free metrics and full-corpus throughput.

# 3 Method

## 3.1 Problem setup

Given embeddings $X = \{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^{1024}$, we learn a quantized representation $\hat{x}_i = \mathcal{Q}_m(x_i)$ under method $m$, and optionally a dequantized approximation $\tilde{x}_i$. We evaluate each method by storage footprint, end-to-end processing throughput, and neighborhood/geometry preservation.

## 3.2 Intrinsic-dimension diagnostics

Before compression sweeps, TerraBit estimates ID using MLE (Levina–Bickel), TwoNN, and local PCA participation ratio across multiple preprocessing variants (raw, centered, L2-normalized, centered+L2). For robustness, the implementation performs stability runs on random subsamples and a neighborhood-size sweep for MLE. These diagnostics guide expected viable compression regimes.

## 3.3 Quantization family

We evaluate seven methods: `float16`, `fp8`, `int8`, `int4`, `int3`, `int2`, and `binary`.

For int$b$ methods ($b \in \{8, 4, 3, 2\}$), we use per-dimension affine quantization with calibration vectors $s$ (scale) and $z$ (zero-point), followed by fixed-width bit packing for $b < 8$. Binary quantization stores only sign bits (packed to bytes), yielding the strongest compression and the highest distortion.

## 3.4 Fidelity metrics

We report label-free metrics designed for retrieval settings:

- kNN recall at $k \in \{10, 25, 50\}$ for cosine and Euclidean distance.
- Pairwise cosine-structure preservation (Spearman correlation).
- Reconstruction cosine similarity and mean-squared error when dequantization is available.

## 3.5 Dataset-preserving transformation pipeline

The `scripts/quantize.py` pipeline processes partitioned Parquet files in parallel. For each file, it reads the embedding column, applies one or more quantizers, replaces only that column, and writes output under per-method roots while preserving directory structure and all non-embedding columns. Quantization metadata is written into Parquet footer metadata under key `quantization`.

# 4 Experiments

## 4.1 Dataset

We evaluate on a large-scale geospatial embedding dataset extracted from a remote sensing foundation model. The dataset consists of 49 785 116 embeddings ($d$=1024, `float32`) stored across 1021 Parquet files totaling approximately 183 GB on disk. Files are Hive-partitioned by geohash prefix (511 level-2 geohash cells), year (2024–2025), and month, with per-file row counts ranging from 174 to 77 856 (mean ≈30 148). Each record carries spatial metadata (geohash, bounding box, WKB geometry), temporal metadata (collection timestamp), and provenance identifiers (chip, cell, raster, STAC item). The embedding column is stored as a variable-length `list<float>` array with uniform length 1024 across all records.

Table 1: **Quantization quality from full-corpus evaluation.** Values are rounded to two decimals. Reconstruction cosine is exact over all rows; C@10/E@10 and cosine correlation are sampled metrics from a 20 000-row reservoir.

| Method | Ratio ($\times$) | Recon. Cos ↑ | C@10 ↑ | E@10 ↑ | Cos Corr ↑ |
|---|---|---|---|---|---|
| float16 | 1.86 | 1.00 | 1.00 | 1.00 | 1.00 |
| fp8 | 3.47 | 1.00 | 0.95 | 0.95 | 1.00 |
| int8 | 3.45 | 1.00 | 0.99 | 0.99 | 1.00 |
| int4 | 6.29 | 1.00 | 0.91 | 0.91 | 1.00 |
| int3 | 7.94 | 0.98 | 0.79 | 0.79 | 0.99 |
| int2 | 11.41 | 0.91 | 0.54 | 0.53 | 0.95 |
| binary | 16.53 | 0.49 | 0.65 | 0.66 | 0.79 |

### 4.2 Protocol

We run two complementary evaluations.

- **Quality regime**: 10 000-sample reservoir subset, quantization-only sweep, label-free fidelity metrics.
- **Systems regime**: full-corpus transformation (49 785 116 rows, 1021 files) over all methods with `jobs=8`, reporting runtime and I/O throughput.

On the quality subset, centered-MLE ID is 17.0 (TwoNN 12.6, L2-centered MLE 20.0), suggesting substantial structural redundancy relative to ambient dimension $d = 1024$.

### 4.3 Quality–compression frontier

Table 1 summarizes quality metrics from the full metrics run (all 1021 files), where reconstruction terms are exact over all rows and neighborhood/correlation terms are computed from reservoir sampling. `int8` and `fp8` retain near-original neighborhood quality at $\approx 4\times$ compression. `int4` gives a strong middle point ($6.29\times$ on-disk, kNN-C@10 $\approx 0.91$). Binary (1-bit) has the largest distortion but still preserves enough local structure (kNN-C@10 0.65, kNN-E@10 0.66) to be practical as a coarse shortlist representation.

### 4.4 Quantization footprint and throughput

We ran full-dataset quantization over all 1021 Parquet files and all 7 methods (`float16`, `fp8`, `int8`, `int4`, `int3`, `int2`, `binary`) with `jobs=8`.

Table 2 summarizes run-level throughput and per-method storage footprint relative to the original 196 366 021 856-byte `float32` embedding corpus.

### 4.5 Discussion

Three observations stand out. First, the measured ID range (roughly 13–20 depending on estimator and preprocessing) is far below 1024, consistent with aggressive quantization without immediate retrieval collapse. Second, the quality decline is gradual from int8 to int2, enabling application-specific operating points rather than a single "best" method. Third, binary quantization is not quality-preserving in a strict sense, but it is operationally attractive as an extreme-compression stage for first-pass candidate generation.

### 4.6 Limitations

Our current study uses label-free proxies rather than task labels, and quality metrics are reported on a sampled subset for computational practicality. We also do not yet benchmark ANN index build/query latency end-to-end. Future work should add labeled downstream tasks, confidence intervals across multiple seeds, and hardware-specific retrieval benchmarks.

Table 2: **Quantization footprint and throughput on the full dataset.** Top block: end-to-end pipeline throughput for all methods combined. Bottom block: per-method output size and effective compression.

| Metric | Value | Metric | Value |
|---|---|---|---|
| Total rows | 49 785 116 | Total files | 1021 |
| Methods | 7 | Workers | 8 |
| Runtime (s) | 2711.79 | Runtime (min) | 45.2 |
| Rows/s | 18 358.77 | Files/s | 0.3765 |
| Output throughput (MiB/s) | 106.90 | Total output (GiB) | 283.10 |
| **Method** | **Size (GiB)** | **Size vs. original (%)** | **Compression (×)** |
| original (float32) | 182.88 | 100.00 | 1.00 |
| float16 | 98.24 | 53.72 | 1.86 |
| fp8 | 52.70 | 28.82 | 3.47 |
| int8 | 52.96 | 28.96 | 3.45 |
| int4 | 29.08 | 15.90 | 6.29 |
| int3 | 23.02 | 12.59 | 7.94 |
| int2 | 16.02 | 8.76 | 11.41 |
| binary | 11.07 | 6.05 | **16.53** |

## 5 Conclusion

TerraBit provides a practical compression workflow for large geospatial embedding corpora. On real 49.8M-row data, we show that low-bit quantization yields a clear quality-cost frontier, with int4/int8 as strong balanced choices and binary as an extreme-compression option for coarse retrieval stages. The code paths and evaluation protocol are designed to be reproducible and easy to adapt to other embedding lakes.

## References